

The University of Edinburgh
School of Philosophy, Psychology and Language Sciences

Attitudes Towards AI Safety
Investigating the Role of Mind Perception in Influencing Concern for
AI Safety



B195766

B186124, B205785

MA (Hons) Psychology and Business

Supervisor: Dr Matti Wilks

Date of Submission: 21/04/2025

Word count: 6000

Acknowledgements

I would first like to thank my supervisor, Dr Matti Wilks, whose guidance and expertise have been invaluable throughout the course of this research. Thank you for introducing me to the fascinating topic of mind perception and AI safety. I've learned so much throughout this process, and I'm especially grateful for your belief in my ability and for celebrating my progress, which helped build my confidence along the way.

I would also like to acknowledge PhD student Ali Ladak, whose extensive knowledge in the field of AI and mind perception was immensely helpful. Thank you for your thoughtful feedback and constant support.

Needless to say, I must thank my fellow dissertation groupmates, Ingrid and Zoe. It has been a true pleasure to work alongside you on this research. Thank you for reassuring my worries and supporting me through to the very end.

I am also grateful to my dearest friend, Lanting, who, having completed her dissertation the year before, offered invaluable feedback on my work and, as always, gave me endless encouragement.

To my closest friends and psychology buddies since day one, Alice and Barbara, thank you for listening to all my ramblings and for being by my side through moments of stress. I couldn't have gone through this without you.

Finally, to my parents—thank you for always being there for me, and for being the strongest source of support and encouragement in all my endeavours. Your unwavering support and love mean everything to me.

Studying psychology has been an incredible and fruitful journey. When I first took psychology in high school, I never imagined it would lead me here—to completing my undergraduate degree with a dissertation in this field. It has been an invaluable experience of learning and growth, and I'm truly grateful to everyone who has shaped and supported me along the way.

Abstract

With the rapid development and increasing adoption of artificial intelligence (AI), experts and policymakers have expressed concerns about the potential for AI to cause large-scale, catastrophic harm, both now and in the future. Public opinion research shows that people often share these concerns, yet the psychological factors that underlie public attitudes towards AI safety remain underexplored. This study draws on research in mind perception to examine how perceived mind capacities in AI relate to concern for AI safety. Specifically, it investigates whether perceived levels of agency and experience in both current and future AI predict concern for AI safety. The study ($N = 215$) used a cross-sectional online survey that measured perceptions of mind capacities, adopted from Gray et al.'s (2007), and concern for AI safety, adopted from Wilks et al. (In prep). A multiple linear regression was conducted to examine the relationships between mind perception and AI safety concern, controlling for demographic variables including age, gender, political orientation, and education. Results showed that higher perceived agency in current AI significantly predicted lower concern for AI safety. In addition, higher perceived experience in future AI significantly predicted higher concern. Perceived agency in future AI and experience in current AI was not significantly associated with concern for AI safety. These findings contribute to theoretical discussions around AI mind perception by offering novel insight into how specific dimensions of perceived mind predict attitudes towards AI safety. Understanding this relationship may support more effective communication, policymaking, and design decisions in the development and deployment of AI.

Introduction

The landscape of AI safety concern

Artificial intelligence (AI) is increasingly embedded in our everyday life, transforming industries through technologies ranging from recommendation algorithms to autonomous systems in customer service and healthcare (Rashid & Kausik, 2024). Broadly, AI refers to computer systems designed to perform tasks that typically require human intelligence, such as reasoning, learning, and language understanding (Russell et al., 2021). However, the rapid and often unpredictable pace of AI development has raised growing concerns about its safety. While there is no universally agreed definition, the UK government's AI Safety Summit broadly defines AI safety as the prevention of harm caused by AI, whether affecting individuals, groups, or society at large (*AI Safety Summit*, n.d.). Scholars have similarly warned that as AI becomes more powerful, the risks it poses can extend beyond data security and privacy concerns, leading to large-scale disruptions, such as labour exploitation or even existential threats to humanity (Carlsmith, 2022; Shulman & Bostrom, 2021).

Public attitudes reflect these anxieties: 59% of Britons express worry about AI, with key concerns including overdependence and the loss of human skills (42%), ethics and potential misuse (37%), and uncertainty around AI's long-term societal impact (33%) (Hooson, 2025). In addition, experts and policymakers are preparing for an AI-integrated future—in 2023, over 350 AI experts and public figures signed a statement urging global prioritisation of efforts to mitigate extinction-level risks posed by AI (*Statement on AI Risk* | CAIS, n.d.). Governments have also begun to act: the UK launched its AI Safety Institute (*AI Safety Institute*, 2024), and California introduced the Safe and Secure Innovation for Frontier AI Models Act (Roth & Davis, 2024).

While these responses reflect a growing recognition of AI risks, without a deeper understanding of the factors influencing public attitudes towards AI safety, we are unable to fully anticipate how society will engage with AI. Psychological mechanisms, such as how people perceive AI, may influence both its adoption and resistance, yet remain underexplored (De Freitas et al., 2023). Gaining insight into these dynamics is crucial for addressing public concerns and guiding ethical AI development that aligns with how people think and feel about AI.

Perceiving the mind of AI

One psychological factor that may influence attitudes towards AI safety, particularly concern about potential AI risks, is mind perception—the tendency to attribute mental states such as thoughts, intentions, or emotions to other entities (Gray et al., 2007). Perceiving others' minds helps people interpret behaviour and make sense of the world (Epley & Waytz, 2010). Hence, exploring the perception of mind in AI may shape understanding on how people interpret and perceive concerns about AI safety. It should be noted that much of the existing literature on mind perception covers a broad range of artificial entities, such as robots, machines, and technological agents. In the present study, these terms will be interpreted as referring to AI, given their shared relevance in understanding how people attribute mental states to artificial intelligence systems.

Dimensions of mind

The influential framework by Gray et al. (2007) proposes that mind perception is structured along two key dimensions: agency and experience (see also Weisman et al., 2017; Malle, 2019, for alternative models of mind). Agency refers to the capacity for intention, self-control, and goal-directed behaviour (Shank et al., 2021). When an entity is perceived as agentic, its actions are interpreted as purposeful, deliberate, and potentially moral (Epley & Waytz,

2010). Experience, on the other hand, refers to the capacity to feel and undergo affective states (Waytz et al., 2010). Entities attributed with experience are thought to possess internal sensations such as pain and hunger, as well as emotions like anger and joy (McMurtrie, 2023). Rather than forming a strict dichotomy, agency and experience exist on a continuum, with entities perceived as having varying degrees of both (Epley & Waytz, 2010). These dimensions are also perceived independently; for example, adults are seen as high in both, God as high in agency but low in experience, and babies as low in agency but high in experience (Gray et al., 2007; McMurtrie, 2023).

When do we see AI as having minds?

In recent years, advances in AI have produced systems that are increasingly autonomous, capable of recognising social cues, and even imitating human actions (Grossberg, 2020; Formosa, 2021). These abilities can make AI *appear* to possess both agency and experience (Geiselmann et al., 2023; Legaspi et al., 2024). However, despite these developments, people still tend to see AI as lacking a full “mind” compared to humans (Shank et al., 2021).

Research suggests that people are generally willing to attribute some agency to AI. For example, Gray et al. (2007) found that robots were perceived as having agency comparable to a young child, and more so than a chimpanzee. AI is also often viewed as capable of making independent decisions and demonstrating self-control (Xu et al., 2021). This is evident in van der Woerdt & Haselager (2019), who found that when a robot appeared to show little effort, people interpreted this as intentional and felt more disappointed by its failure. This suggests that AI is perceived as agentic when its actions appear deliberate, reinforcing the belief that it exercises some control over its behaviour rather than functioning merely as a passive tool.

On the other hand, people are averse to attributing experience to AI. Robots are often seen as having experiential capacities comparable to a dead person (Gray et al., 2007) and are thought to experience pleasure or pain no more than a rock (Ladak et al., 2025). However, in some cases, people are willing to grant AI experiential capacities. For instance, Wang & Krumhuber (2018) found that social robots that were designed for companionship were perceived as more emotionally capable than robots built purely for economic purposes. Similarly, Tanibe et al. (2017) found that participants who had helped a robot were more likely to believe it could feel pleasure. These findings suggest that people may sometimes attribute experience to AI, especially in social or emotionally engaging contexts.

How do people feel about AI having minds?

Although people may attribute AI some agency, they are generally uncomfortable with highly agentic AI, especially when it makes moral decisions, such as performing high-risk surgeries, or operating independently in sensitive domains such as law, the military, or autonomous vehicles (Bigman & Gray, 2018; Young & Monroe, 2019). The idea that AI could genuinely have experience tends to provoke even stronger discomfort (Gawood, 2003). This aligns with the uncanny valley effect, where entities that appear almost human but not fully so can evoke unease (Gray & Wegner, 2012). This may be because people do not expect robots to feel emotions or possess internal states, so AI that is capable of doing so can seem unsettling (Mori et al., 2012). Given this general discomfort around AI minds, it is intriguing to explore how such perceptions influence concern for AI safety.

The impact of perceived mind on perceptions of AI safety

Research indicates that perceptions of mind in AI influence trust, blame attribution, and perceived threat (Bigman et al., 2022; Gogoll & Uhl, 2018). For instance, higher perceived agency in AI has been linked to increased uncertainty (Liu, 2021), reduced reuse intentions (Stafford et al., 2013), and sharper declines in trust (Vanneste & Puranam, 2024). The following sections illustrate how perceptions of mind in AI relate to blame and perceived threat, and how these perceptions contribute to broader AI safety concern.

Blame attribution

Research indicates that when AI is perceived as agentic, its actions are more likely to be interpreted as intentional and, consequently, judged more harshly (Waytz et al., 2014; Shank and DeSanti, 2018). For instance, when an algorithm responsible for a brand crisis was described as capable of thinking, planning, and exercising moral judgment similar to a human, people perceived the incident as more intentional and attributed greater blame than when the algorithm lacked such capacities (Srinivasan and Sarial-Abi, 2021). This tendency to assign blame reflects a heightened sensitivity to the idea that the harm was avoidable, and that AI had the capacity to act differently but failed to do so (Lagnado and Channon, 2008; Malle et al., 2014). In this sense, attributing agency to AI not only increases blame but also amplifies perceptions of its potential to cause deliberate harm, thereby fuelling concerns about its potential to cause larger-scale disruptions related to AI safety.

In addition, perceived mind in AI influences how much responsibility is attributed to it (Joo, 2024). People are more likely to shift blame from corporations to AI and hold it accountable for ethical violations when the AI involved is described as capable of sophisticated

thought and understanding of human emotions (Kneer & Stuart, 2021; Joo, 2024). This aligns with the concept of blame attribution asymmetry, where people tend to assign greater responsibility and harsher blame to autonomous AI than to humans, even when the harm is identical (Liu & Du, 2021). The researchers argue that this asymmetry may stem from heightened fear and anxiety triggered by AI-caused harm. As AI is perceived to have more agency, people expect it to assume more responsibility, leading to intensified emotional reactions in response to negative outcomes. Together, these attributions of blame suggest that perceiving agency in AI may intensify concerns about its capacity for harm and contribute to anxiety regarding AI safety (Kim et al., 2025; Mannes, 2020).

Threat to distinctiveness

Perceiving AI as having a mind can further increase concerns about its threat to human distinctiveness, as the mind is often regarded as uniquely human and central to our identity (Huebner, 2010). While animals are also seen as having minds, they are not viewed as threatening and are more readily forgiven for causing harm than AI (Chessman, 2018). This may be because humans perceive themselves as mentally superior and see animals as having simpler minds (Haslam et al., 2008). In contrast, AI with human-like minds evokes discomfort by triggering perceptions of danger and competition (Dietvorst et al., 2015; Frank et al., 2019). Ferrari et al. (2016) termed this the threat to distinctiveness hypothesis, which suggests that blurring the boundaries between humans and machines creates a sense of threat. Supporting this, studies show that while robots may be accepted for their accuracy or efficiency, their display of experiential qualities, such as determination or compassion, can be unsettling because it diminishes clear distinctions from humans (Giger et al., 2024; Müller, 2020). Moreover, this perceived threat is found to extend beyond the identity domain (e.g., “Technological

advancement in the area of robotics is threatening to human uniqueness.”) to reality (e.g., “In the long run, robots pose a direct threat to human safety and wellbeing.”) (Złotowski et al., 2017). This suggests that perceiving AI as having a mind not only evokes fears of competition but also heightens concerns about its potential to interfere with human life and pose safety risks.

The Present Study

The present study offers an initial exploration into the relationship between the perception of mind in AI and attitudes towards AI safety. While previous studies investigated how mind perception in AI influences blame and threat (Liu & Du, 2021; Ferrari et al., 2016), we extend the literature by linking perceptions of agency and experience to broader safety concerns. Furthermore, the present study distinguishes between current and future AI, recognising that people often hold different expectations for future AI compared to the current AI they interact with. For instance, future AI is more readily attributed emotional capacities (Pauketat & Anthis, 2022), possibly because it is expected to achieve deeper societal integration and more radical advancements (Shabbir & Anwer, 2018). As a result, future AI may be seen as more threatening than current AI, offering important insights into safety concerns. Beyond its theoretical contribution, the study has practical relevance: its findings may inform the design of decision-making systems, such as autonomous vehicles and intelligent software, in ways that may improve public acceptance and usability. Additionally, the study may support evidence-based policymaking and contribute to the ethical development and regulation of AI technologies. The following hypotheses were preregistered and tested:

H1a Higher perceived agency in current AI is associated with greater concern for AI safety.

H1b Higher perceived agency in future AI is associated with greater concern for AI safety.

H2a Higher perceived experience in current AI is associated with greater concern for AI safety.

H2b Higher perceived experience in future AI is associated with greater concern for AI safety.

Methodology

The study was approved by the University of Edinburgh PPLS Research Ethics Committee (81-2425/3). The study hypotheses and methods were preregistered; the full preregistration form can be found on <https://aspredicted.org/ntxz-8sc9.pdf>.

Participants

A-priori power analysis indicated that, to detect a medium effect ($f = .27$) with a power of .80, the study required a sample of at least 165 participants. To account for potential attrition, we aimed for an additional 20%, resulting in a target sample size of 198. A total of 274 participants were recruited, of these, 158 were recruited via The University of Edinburgh's undergraduate participant pool (their participation was compensated with 0.5 credits), the rest were via snowball sampling through the research group member's social networks on social media. The final sample consisted of 215 participants. This number was reached after excluding participants who did not consent ($n = 2$), failed to complete either of the two attention checks ($n = 45$), were underage ($n = 5$), or did not complete all the questions necessary for analysis ($n = 41$). Of these participants, age ranged from 18 to 56 ($M = 22.29$, $SD = 8.62$), with 73.5% identifying as female, 21.4% identifying as male, and 5.1% identifying as other. The sample was moderately liberal, with a mean score of 3.38 on a 9-point scale, where higher values represent a more conservative orientation. The most common education level among participants was a high school diploma ($n = 121$), and the majority identified as either White (48.8%) or Asian (38.6%). Full sample demographic measures are provided in Appendix A.

Design

The study used a cross-sectional online survey as the primary data collection tool. Participants answered a series of Likert-scale questions that quantified their perceptions on AI mental capacities and AI safety. To minimise order effects, all questions and measures were presented in randomised order.

Materials

AI safety concern

Concern for AI safety was measured using five items adopted from Wilks et al. (In prep). These items were chosen due to their strong relevance to AI safety concerns and showed an acceptable internal reliability (*Cronbach's* $\alpha = .77$). For example, one item stated: "*The development of highly advanced AI could be one of the greatest threats to life on Earth.*" Participants rated their agreement with each statement on a 7-point Likert scale (1 = strongly disagree, 7 = strongly agree). Higher scores indicated a greater concern for AI safety. Full AI safety measures are provided in Appendix B.

Mind perception

Mind perception was measured using 10 items adopted from Gray et al.'s (2007) mind dimensions framework. Five items represented agency (e.g., self-control), and five items represented experience (e.g., hunger), selected based on their highest factor loadings. The full list of item loadings from Gray et al. (2007) is provided in Appendix C. Both subscales demonstrated strong internal reliability (agency: *Cronbach's* $\alpha = .83$; experience: *Cronbach's* $\alpha = .91$). Participants rated the extent to which they believed AI systems possess each capacity on a 7-point Likert scale (1 = strongly disagree, 7 = strongly agree). Higher scores reflected a greater

attribution of the relevant mental capacity to AI. To examine whether perceptions of mind in future AI differ from current AI in their relationship with AI safety concern, participants completed two versions of the mind perception measure: one referring to current AI and the other future AI. Full mind perception measures are provided in Appendix D.

Covariates

Demographics were included to control for individual differences that could influence perceptions of AI (Kaya et al., 2022). This included age, gender, political orientation (PO), and education level.

Procedure

The survey was conducted online via Qualtrics. After providing informed consent, participants first read a definition of AI derived from Pauketat & Anthis (2022) (see Appendix B). Then, participants completed the question measuring AI safety concern, were reminded of the definition, and responded to the questions on AI mind perception. They also responded to measures of construal level, perceived capabilities of current and future AI, conspiratorial ideation, and social dominance orientation; these measures were part of a larger project and are not included in the present study. To further ensure the quality of responses, two attention checks were incorporated throughout the survey (see Appendix E). Finally, participants provided demographic information. They were thanked and debriefed at the end of the survey.

Planned data analysis

All statistical analyses will be conducted in R (version 4.3.3, 2024; R Core Team, 2024). To check for potential collinearity, correlations between the independent variables will be

examined prior to the main analyses using the *stats* package (version 4.2.3; R Core Team, 2024). Correlations between individual capacities will also be explored. The primary analysis will use multiple linear regression, also conducted with the *stats* package (version 4.2.3; R Core Team, 2024). A multiple linear regression model was chosen as the data were continuous and the relationships between variables were expected to be linear. The dependent variable, AI safety concern, will be calculated as the mean score across five items. Independent variables will include mean scores for current AI agency, current AI experience, future AI agency, and future AI experience, each computed as the average of their respective sets of five capacity items. Age, gender, PO, and education will be included as covariates. Results will be considered statistically significant at $\alpha = .05$. In cases where predictors are found to be highly correlated, they will be analysed in separate models to mitigate multicollinearity.

Results

All assumptions of linearity, independence of errors, homoscedasticity, and normality of residuals were visually assessed and met. Full diagnostic details are provided in Appendix F.

As part of the preliminary analysis, Pearson pairwise correlations were computed between variables to examine collinearity and explore relationships among key constructs (see Table 1). Most correlations between the independent variables, though statistically significant, were weak to moderate. The strongest relationship was between perceived current agency and future agency of AI, suggesting that those who perceive more agency in current AI are likely to anticipate greater agency in future AI. Nevertheless, no correlation coefficients exceeded .70, suggesting that multicollinearity was unlikely to pose a major issue. This was further supported by VIF values in later analysis ($VIF < 5$), indicating that multicollinearity did not adversely affect model estimates (see Appendix F).

Correlations between the independent variables and AI safety concern revealed no significant relationships, except for a weak negative correlation between concern for AI safety and the perception of current AI agency ($r = -.18, p < .01$). This suggests that as participants perceive current AI as more agentic, their concern for AI safety slightly decreases. These relationships were subsequently examined using multiple linear regression analyses.

Table 1*Pearson Pairwise correlations between the independent variables and AI safety concern*

	Agency (Current)	Agency (Future)	Experience (Current)	Experience (Future)	Age	PO
Agency (Current)						
Agency (Future)	0.70***					
Experience (Current)	0.32***	0.14*				
Experience (Future)	0.43***	0.42***	0.68***			
Age	0.01	-0.06	0.29***	0.17*		
PO	-0.00	0.04	0.17*	0.06	0.08	
AI Safety Concern	-0.18**	-0.05	-0.01	0.10	0.09	-0.03

Note. * $p < .05$, ** $p < .01$, *** $p < .001$.

In addition, correlations between mental capacities within and between agency and experience were examined (see Tables 2 and 3). For both current and future AI, capacities within each mind dimension showed weak to moderate positive correlations, with generally stronger and more significant associations for future AI. Notably, within the agency dimension, morality was weakly to moderately correlated with all other agentic capacities, particularly for future AI. This suggests that to some extent, participants considered AI systems as having the capacity for moral judgement alongside agentic capacities such as self-control (current: $r = .30, p < .001$; future: $r = .47, p < .001$), and emotion recognition (current: $r = .40, p < .001$; future: $r = .42, p < .001$). Experiential capacities also showed weak to moderate correlations with some agentic capacities, like morality, self-control, and emotion recognition.

Table 2*Pearson Pairwise correlations between individual mental capacities in **current AI***

	Agency					Experience			
	Self-control	Morality	Memory	Emotion Recognition	Planning	Hunger	Fear	Pain	Pleasure
Self-control									
Morality	0.30***								
Memory	0.20**	0.16*							
Emotion Recognition	0.21**	0.40***	0.22**						
Planning	0.27***	0.24***	0.49***	0.21**					
Hunger	0.12	0.22**	-0.04	0.07	-0.02				
Fear	0.22***	0.31***	-0.02	0.16*	0.11	0.46***			
Pain	0.21**	0.27***	0.00	0.20**	0.09	0.55***	0.59***		
Pleasure	0.16*	0.38***	0.07	0.27***	0.11	0.43***	0.52***	0.49***	
Rage	0.22***	0.21**	0.08	0.24***	0.05	0.43***	0.58***	0.51***	0.61***

Note. * $p < .05$, ** $p < .01$, *** $p < .001$.

Table 3*Pairwise Pearson correlations between individual mental capacities in **future AI***

	Agency					Experience			
	Self-control	Morality	Memory	Emotion Recognition	Planning	Hunger	Fear	Pain	Pleasure
Self-control									
Morality	0.47***								
Memory	0.31***	0.24***							
Emotion Recognition	0.35***	0.42***	0.34***						
Planning	0.30***	0.22**	0.62***	0.48***					
Hunger	0.22***	0.25***	0.01	0.26***	0.06				
Fear	0.32***	0.40***	0.06	0.31***	0.12	0.56***			
Pain	0.27***	0.32***	0.04	0.25***	0.10	0.72***	0.73***		
Pleasure	0.28***	0.38***	0.17*	0.27***	0.21**	0.58***	0.61***	0.64***	
Rage	0.36***	0.31***	0.12	0.32***	0.17*	0.54***	0.62***	0.58***	0.63***

Note. * $p < .05$, ** $p < .01$, *** $p < .001$.

Descriptive statistics are presented in Table 4. On average, participants rated perceptions of agency in current AI higher ($M = 4.56$, $SD = 1.03$) than perceptions of experience ($M = 2.13$, $SD = 1.13$), a pattern that was also observed in future AI. This suggests that, regardless of time

frame, AI is generally seen as having more capacity for deliberate action rather than for experiencing or feeling. Additionally, participants attributed higher agency and experience to future AI than to current AI. This indicates that participants anticipate more advanced minds in future AI, though there is greater variation in these perceptions. Regarding AI safety, participants expressed moderate to high concern, with scores ranging from 2.2 to 7 ($M = 5.20$, $SD = 1.02$). Overall, there was notable variation in participants' responses, reflecting differing levels of concern and attributions of mental capacities.

Table 4

Descriptive statistics of independent and dependent variables

Independent Variable	Current			Future		
	Mean	SD	Range	Mean	SD	Range
Agency^a	4.56	1.03	1 - 6.8	5.41	1.10	1 - 7
Experience^a	2.13	1.13	1 - 5.4	3.01	1.57	1 - 7
Dependent Variable				Mean	SD	Range
AI Safety Concern^a				5.2	1.02	2.2 - 7

Note. SD = standard deviation. ^a Max range 1 – 7.

A multiple linear regression was conducted to examine how perceptions of agency and experience in current and future AI relate to concern for AI safety. The model was statistically significant ($F(13,201) = 2.99, p < .001$), accounting for approximately 16.2% of the total variance in AI safety concern scores ($R^2 = .162$; adjusted $R^2 = .108$) (see Table 5).

Contrary to the prediction in Hypothesis 1a, higher perceived agency in current AI significantly predicted *lower* concern ($\beta = -0.34, SE = 0.10, p = .001$), indicating that participants who attributed more agency to current AI tend to express less concern for AI safety. No significant effect was found for perceived agency in future AI ($\beta = 0.08, SE = 0.09, p = .366$).

In line with Hypothesis 2b, higher perceived experience in future AI significantly predicted *greater* concern for AI safety ($\beta = 0.19$, $SE = 0.06$, $p = .003$). However, perceived experience in current AI did not significantly predict concern ($\beta = -0.13$, $SE = 0.09$, $p = .146$). Figure 1 visually represents these relationships, showing how concern for AI safety varies with perceptions of agency and experience across both current and future AI.

Among the demographic covariates, no significant effects were observed except for a gender subgroup ($\beta = -0.81$, $SE = 0.34$, $p = .017$); however, this result should be interpreted with caution, as only 11 participants (5% of the sample) identified as “other.” This small subgroup size suggests the effect may be due to chance or sampling variability, with a negligible impact on concern for AI safety.

Table 5

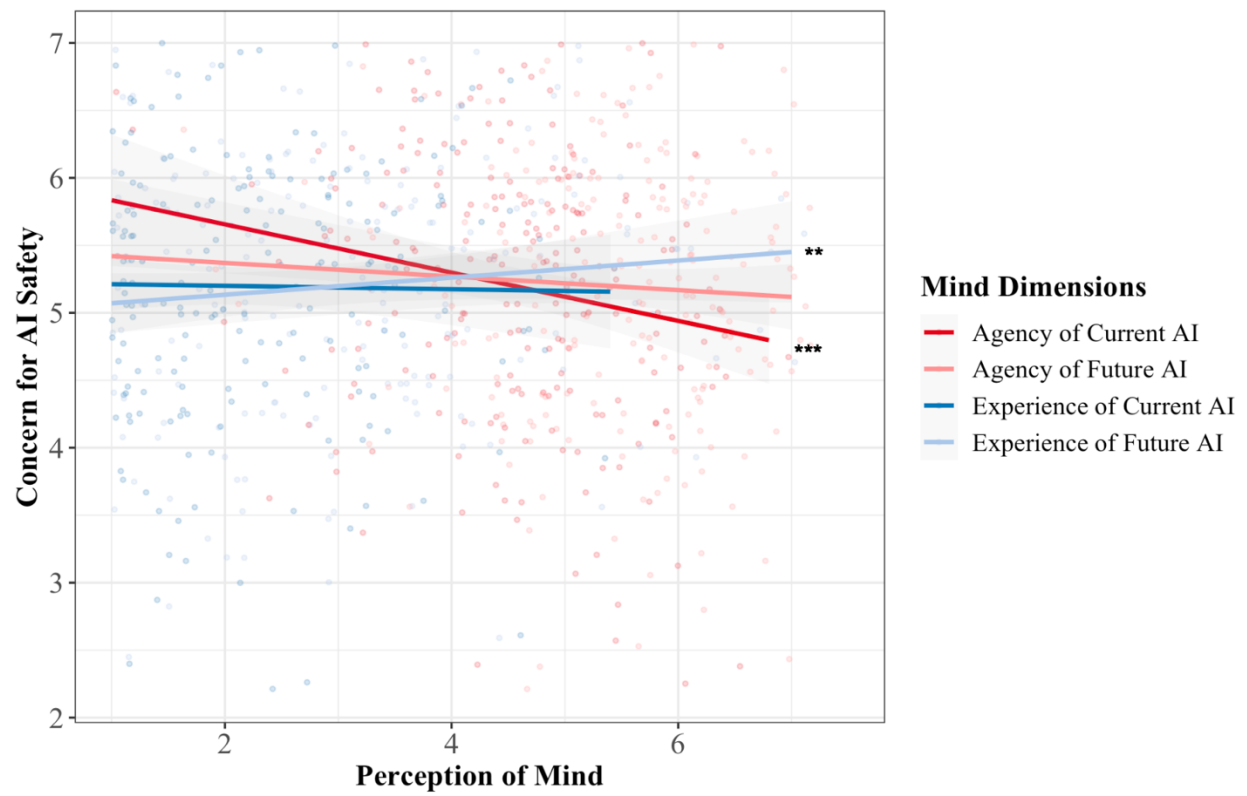
Multiple linear regression model for mental capacities in current and future AI on concern for AI safety

<i>Predictors</i>	Concern for AI Safety			
	<i>Estimates</i>	<i>std. Error</i>	<i>CI</i>	<i>p</i>
Intercept	5.62	0.48	4.67 – 6.57	<0.001
Agency_Current AI	-0.34	0.10	-0.53 – -0.15	0.001
Agency_Future AI	0.08	0.09	-0.10 – 0.27	0.366
Experience_Current AI	-0.13	0.09	-0.30 – 0.05	0.146
Experience_Future AI	0.19	0.06	0.07 – 0.32	0.003
Age	0.02	0.01	-0.00 – 0.04	0.050
Gender (Woman)	0.26	0.17	-0.08 – 0.59	0.130
Gender (Other)	-0.81	0.34	-1.48 – -0.15	0.017
Political Orientation	-0.05	0.04	-0.13 – 0.04	0.292
Education (Some College)	-0.19	0.17	-0.54 – 0.15	0.265
Education (Associate)	-1.05	0.71	-2.44 – 0.34	0.139
Education (Bachelor's)	0.02	0.21	-0.39 – 0.43	0.929
Education (Master's)	-0.21	0.32	-0.84 – 0.41	0.504
Education (Professional/Doctorate)	-0.17	0.60	-1.34 – 1.00	0.775
Observations	215			
R ² / R ² adjusted	0.162 / 0.108			

Note. *CI* = confidence interval.

Figure 1

Associations between perception of mind in current and future AI and concern for AI safety



Note. * $p < .05$, ** $p < .01$, *** $p < .001$.

Discussion

The present study explored the relationship between perception of mind in AI and attitudes towards AI safety. Specifically, we tested whether higher perceived agency and experience in AI predicted greater concern for AI safety for both current and future AI. Supporting Hypothesis 2b, higher perceived experience in future AI predicted greater concern for AI safety. However, contrary to Hypothesis 1a, higher perceived agency in current AI was associated with lower safety concern. Our findings also did not support Hypothesis 1b, which posited that higher perceived agency in future AI would predict greater concern, nor Hypothesis 2a, which posited that higher perceived experience in current AI would predict greater concern. The following sections discuss these findings in relation to current literature, offering possible explanations and exploring their implications for understanding concern for AI safety.

The impact of perceived agency on perceptions of AI safety

The findings indicate that people are willing to attribute a moderate level of agency to AI, especially to future AI. Notably, perceived agency appears to be related to reduced concern for AI safety.

Agency in current AI

Our finding in relation to Hypothesis 1a was unexpected, given prior research suggesting that attributing agency to AI increases perceptions of responsibility and blameworthiness, thereby heightening concerns about AI risks (Waytz et al., 2014; Ayad & Plaks, 2025; Srinivasan & Sarial-Abi, 2021). However, this result aligns with other studies indicating that perceived agency in AI can foster trust, particularly when it demonstrates competence and reliability (Troshani et al., 2020; Madhavan & Wiegmann, 2007). For instance, Lee et al. (2020) found that

participants who perceived chatbots as having higher agency, such as the ability of autonomous conversation and conscious action, reported greater closeness and willingness to use them when outcomes were satisfying (i.e., receiving recommendations aligned with personal interests). Likewise, people were more willing to accept machine decisions when its expertise was made salient (Bigman & Gray, 2018). This suggests that perceived agency in AI increases trust, particularly in contexts where AI is expected to succeed or has succeeded, which contrasts research linking perceived agency to blame, typically in cases involving errors or harm. Supporting this, Schoenherr and Thomson (2024) highlight the importance of context, finding that autonomous AI was trusted more than humans when successful but significantly distrusted when it failed. Together, these findings imply that trust in agentic AI increases not only when it is seen as competent, but also when it delivers successful outcomes.

Therefore, a related explanation for our result is that when prompted with current AI, people may have drawn on familiar examples of AI they interact with, that tend to be competent, reliable, and have caused little observable harm in everyday use. As a result, they may see less reason to be concerned. This real-world familiarity fosters more positive attitudes towards AI, which may be linked to lower safety concerns (Horowitz et al., 2024). However, this interpretation remains tentative. Despite higher perceived agency in current AI was associated with lower concern, overall concern for AI safety remained moderate to high. This suggests that although familiarity with competent and successful AI may have reduced some concern, broader anxieties about AI safety persist, which should be explored further.

Agency in future AI

No significant effect was found for Hypothesis 1b. One explanation, in relation to the interpretation of current AI agency, is that participants felt less confident linking future AI agency to successful outcomes, as some researchers suggest that AI failures will become more frequent and severe in the future (Yampolskiy, 2019). However, this perception may not have been strong enough within our sample to produce a significant effect.

Another possible explanation is that participants associated agentic capacities, such as planning and self-control, with morality more in future AI. Research shows that highly competent AI systems lacking a moral framework are seen as more dangerous (Floridi et al., 2018). For example, Borenstein et al. (2017) found that concerns about autonomous vehicles arise when these systems, though highly competent in navigation, lack the moral decision-making necessary for life-and-death situations. The fear is that AI systems without moral consideration could make decisions misaligned with human values, leading to catastrophic outcomes (Amodei et al., 2016; Binns et al., 2018). In our study, weak positive correlations between morality and other agentic capacities in future AI suggest that people expect future AI to be not only competent and self-aware, but also capable of moral reasoning. This expectation may have mitigated concerns typically associated with higher perceived agency in future AI, potentially explaining why safety concerns did not increase. However, the limited existing literature and the lack of replication warrant further investigation into these relationships.

The impact of perceived experience on perceptions of AI safety

The results show that people are willing to attribute limited experience to AI, more so for future than current AI. However, while people may be comfortable with AI possessing some degree of experience, higher perceptions is associated with increased safety concern.

Experience in current AI

The lack of a significant relationship between perceived experience in current AI and safety concerns may suggest that people do not see AI as possessing sufficient experience to pose a real threat. Current AI is still predominantly regarded as a tool for calculation and problem-solving, rather than as an entity with emotions or subjective experience (Ladak et al., 2023). Although recent advancements have increased the complexity of AI, the extent to which they possess experience or consciousness remains widely debated (Hildt, 2019). This implies that, at present, perceived experience in current AI is not strongly relevant to safety concerns, though it may become more salient as AI systems begin to exhibit more experiential traits.

Experience in future AI

In line with our prediction, higher perceived experience in future AI was associated with higher concern for AI safety. This supports prior research demonstrating that overly human-like qualities in artificial agents can elicit feelings of uncanniness and reduce trust (Sullivan et al., 2022). For example, digital doppelgangers with realistic, experiential features have been found to evoke discomfort (Weisman & Peña, 2021), and robots with emotions are thought to be less trustworthy (Nijssen et al., 2022).

Furthermore, this finding may play a role in supporting Ferrari et al. (2016)'s threat to distinctiveness hypothesis, in confirming that people feel threatened when AI is attributed experiential capacities like pain and hunger, which are seen as inherently human. The blurring of psychological boundaries between human and AI may have provoked concerns about AI as an existential risk. Philosophically, researchers argue that if people believe future AI can have subjective experiences like humans, they may also infer it could possess desires, preferences, and

the ability to manipulate human emotional responses (Lynch, 2021; Corbeel, 2024). These attributes may fuel concerns that experiential AI could eventually pursue goals misaligned with human interests and values, heightening fears about its capacity for harm and societal disruption (Yudkowsky, 2008). However, these concerns remain speculative and require empirical investigation into whether people actually hold such worries.

Conversely, some studies have found contradicting evidence, in which highlighting AI's experience has led to more positive attitudes (Bigman & Gray, 2018; Young & Monroe, 2019). In particular, the attribution of experience may influence perceptions of AI as moral patients—entities capable of being helped or harmed, and thus deserving of moral consideration and empathy (Ladak et al., 2023). For instance, robots that were attributed greater capacity for experiencing and expressing emotions were less likely to be harmed (Wang & Krumhuber, 2018) and more accepted in its use (Bigman & Gray, 2018). Similar effects were observed in reduced blame, anger, and intentions to discard experiential AI when they committed moral violations (Tanibe et al., 2017). These findings indicate that enhancing perceptions of experience in AI can reduce aversion and increase acceptance, which contrasts with our finding. Given these mixed results, further research is needed to clarify how perceived experience in AI shapes attitudes toward AI safety.

Limitations and Future Directions

The present study has several limitations that should be considered. One methodological limitation is the use of only ten capacities from Gray et al.'s (2007) 18-factor mind perception framework. Including a broader range of items could provide a more robust understanding of how people perceive AI minds. Additionally, this study adopted a single conceptualisation of

mind perception. While we made clear distinctions between agency and experience, their respective mental capacities often overlap in other mind perception frameworks. For example, Weisman et al.'s (2017) Mind–Body–Heart model examines how people organise minds into distinct trait clusters, in which concern for AI safety may arise from combinations of perceived mental traits that feel particularly unsettling or threatening. Hence, perceptions of mind measured through different conceptualisations may present different insights into safety concerns. Finally, the AI safety concern measure, which focused on existential risk and personal concern, may not fully capture concepts like threat to distinctiveness or blame. While our results suggest some relevance to these ideas, aligning measures more closely with these concepts may provide better assessments on their relationships.

It is also important to consider the potential conceptual ambiguity between perceived *capacity* (i.e., the ability to hold internal traits) and perceived *capability* (i.e., the ability to demonstrate power or competence) (Vincent & Gannon, 2008). While participants were explicitly asked to rate mental capacities, the safety concern items (e.g., “the development of advanced AI”) may have prompted them to think about what AI is practically capable of doing. The blurring of concepts may have contributed to our finding that higher perceived agency in current AI predicted lower safety concern, as perceptions of greater capability are often associated with greater comfortableness with AI (Schepman & Rodway, 2020). Moreover, the null result for future AI agency could have been due to greater uncertainty associated with future AI capabilities. These explanations imply that the conceptual overlap between AI capacity and capability may have influenced the outcomes. Future studies should aim to distinguish and measure these constructs separately to better assess their roles in predicting AI safety concern.

Another limitation is the broad conceptualisation of AI in this study, where we referred to AI in general terms. However, research indicates that when asked about AI, people form diverse mental representations of AI's physical embodiments (Hwang & Won, 2022), which can vary significantly in perceived levels of agency and experience (Ladak et al., 2025). For example, socially oriented or lifelike AI (e.g., robotic companion animals) is often seen as high in experience but low in agency, while automative AI (e.g., self-driving cars) is seen as only high in agency, and utilitarian AI (e.g., robot vacuum cleaners) as low in both (Ladak et al., 2025; Jacobs et al., 2022). These distinctions are likely more pronounced for current AI, as people are more likely to base their perceptions on direct experiences with different types of AI. For example, when asked about current AI, a participant imagining a Tesla self-driving car would perceive greater agency than one imagining a robot vacuum cleaner. This implies that our broad framing may have overlooked key nuances in AI mind perception. Future research should consider how different types of AI shape perceptions of mind and subsequent effects on safety concerns.

In addition, our sample was predominantly liberal, female, and composed mainly of students residing in the UK, with limited ethnic diversity. These factors may limit the generalisability of the findings. For instance, culture shapes people's conceptualisations of AI: while individualistic cultures are more likely to view AI as a distinct and potentially threatening agent, collectivistic cultures tend to perceive it as more harmonious and integrated (Barnes et al., 2024). Follow-up studies are needed to examine how such differences may influence varying levels of concern for AI safety.

Finally, beyond mind perception, there are a range of psychological and social factors, such as exposure to AI, sci-fi fan identity or beliefs like substratism (the denial of mental states to non-biological entities; see Anthis et al., 2025), that may collectively influence judgements

and perceptions of AI. For instance, those who are more exposed to AI tend to attribute it greater agency but less experience (Xu & Sar, 2018). However, other studies suggest that exposure to AI can increase beliefs that it can experience distress (Vuong et al., 2023) and reduce the uncanny valley effect (Breazeal, 2002). Taking these differences into account could add clarity and insight into our findings and provide a more comprehensive understanding of the various psychological influences shaping attitudes towards AI safety.

Conclusion

Amid growing discussions around AI's societal impact, especially its potential to pose safety risks, understanding the psychological foundations of public concern has become increasingly important. This study examined how perceptions of mind in AI relate to concern for AI safety. Unexpectedly, greater perceived agency in current AI was associated with lower safety concern, possibly reflecting increased trust due to high perceived competence and reliability of familiar AI. As anticipated, higher perceived experience in future AI was associated with greater concern for AI safety, supporting existing theories that link experiential AI with feelings of uncanniness and perceived threats to human identity. However, as the literature presents evidence supporting both heightened and reduced concern depending on the context, further research is needed to clarify when and how perceptions of experience shape attitudes towards AI safety. No significant associations were found for current AI experience or future AI agency. This could reflect certain limitations of the study or suggest that these variables are not predictive on their own. Despite these nuances, the study contributes to an emerging field by offering a novel perspective on how mind perception influences concern for AI safety. Future research and replication may deepen our understanding of the psychological dynamics underlying public attitudes toward AI.

References

- AI Safety Institute*. (2024, May 22). GOV.UK. <https://www.gov.uk/government/organisations/ai-safety-institute>
- AI Safety Summit: introduction (HTML)*. (n.d.). GOV.UK. <https://www.gov.uk/government/publications/ai-safety-summit-introduction/ai-safety-summit-introduction-html>
- Amodei, D., Olah, C., Steinhardt, J., Christiano, P. F., Schulman, J., & Mané, D. (2016). Concrete Problems in AI Safety. *ArXiv (Cornell University)*. <https://doi.org/10.48550/arxiv.1606.06565>
- Anthis, J. R., Pauketat, J. V. T., Ladak, A., & Manoli, A. (2025). *Perceptions of Sentient AI and Other Digital Minds: Evidence from the AI, Morality, and Sentience (AIMS) Survey*. <https://doi.org/10.1145/3706598.3713329>
- Ayad, R., & Plaks, J. E. (2025). Attributions of intent and moral responsibility to AI agents. *Computers in Human Behavior: Artificial Humans*, 3, 100107. <https://doi.org/10.1016/j.chbah.2024.100107>
- Barnes, A. J., Zhang, Y., & Valenzuela, A. (2024). AI and culture: Culturally dependent responses to AI systems. *Current Opinion in Psychology*, 58, 101838. <https://doi.org/10.1016/j.copsyc.2024.101838>
- Bigman, Y. E., & Gray, K. (2018). People are averse to machines making moral decisions. *Cognition*, 181, 21–34. <https://doi.org/10.1016/j.cognition.2018.08.003>
- Bigman, Y. E., Wilson, D., Arnestad, M. N., Waytz, A., & Gray, K. (2022). Algorithmic discrimination causes less moral outrage than human discrimination. *Journal of Experimental Psychology: General*, 152(1). <https://doi.org/10.1037/xge0001250>

- Binns, R., Van Kleek, M., Veale, M., Lyngs, U., Zhao, J., & Shadbolt, N. (2018). 'It's Reducing a Human Being to a Percentage': Perceptions of Justice in Algorithmic Decisions. *Proceedings of the 2018 CHI Conference on Human Factors in Computing Systems*, 1–14. <https://doi.org/10.1145/3173574.3173951>
- Borenstein, J., Herkert, J., & Miller, K. (2017). Self-Driving Cars: Ethical Responsibilities of Design Engineers. *IEEE Technology and Society Magazine*, 36(2), 67–75. <https://doi.org/10.1109/MTS.2017.2696600>
- Breazeal, C. L. (2002). *Designing sociable robots*. Mit Press.
- Carlsmith, J. M. (2022). Is Power-Seeking AI an Existential Risk? *ArXiv (Cornell University)*. <https://doi.org/10.48550/arxiv.2206.13353>
- Chessman, C. (2018). Not Quite Human: Artificial Intelligence, Animals, and the Regulation of Sentient Property. *SSRN Electronic Journal*. <https://doi.org/10.2139/ssrn.3200802>
- Corbeel, B. (2024, January 1). *Does AI Experience 'Desire', and Can We Analyze It in Terms of Psychoanalytic Drive Theory?* Medium. <https://medium.com/@brehtcorbeel/does-ai-experience-desire-and-can-we-analyze-it-in-terms-of-psychoanalytic-drive-theory-ef867e2b47fc>
- De Freitas, J., Agarwal, S., Schmitt, B., & Haslam, N. (2023). Psychological factors underlying attitudes toward AI tools. *Nature Human Behaviour*, 7(11), 1845–1854. <https://doi.org/10.1038/s41562-023-01734-2>
- Dietvorst, B. J., Simmons, J. P., & Massey, C. (2015). Algorithm aversion: People erroneously avoid algorithms after seeing them err. *Journal of Experimental Psychology: General*, 144(1), 114–126. <https://doi.org/10.1037/xge0000033>

Epley, N., & Waytz, A. (2010). Mind Perception. *Handbook of Social Psychology*.

<https://doi.org/10.1002/9780470561119.socpsy001014>

Ferrari, F., Paladino, M. P., & Jetten, J. (2016). Blurring Human–Machine Distinctions:

Anthropomorphic Appearance in Social Robots as a Threat to Human Distinctiveness.

International Journal of Social Robotics, 8(2), 287–302. <https://doi.org/10.1007/s12369-016-0338-y>

Floridi, L., Cowls, J., Beltrametti, M., Chatila, R., Chazerand, P., Dignum, V., Luetge, C.,

Madelin, R., Pagallo, U., Rossi, F., Schafer, B., Valcke, P., & Vayena, E. (2018).

AI4People—An Ethical Framework for a Good AI Society: Opportunities, Risks, Principles, and Recommendations. *Minds and Machines*, 28(4), 689–707.

<https://doi.org/10.1007/s11023-018-9482-5>

Formosa, P. (2021). Robot Autonomy vs. Human Autonomy: Social Robots, Artificial

Intelligence (AI), and the Nature of Autonomy. *Minds and Machines*, 31(4), 595–616.

<https://doi.org/10.1007/s11023-021-09579-2>

Frank, M. R., Autor, D., Bessen, J. E., Brynjolfsson, E., Cebrian, M., Deming, D. J., Feldman,

M., Groh, M., Lobo, J., Moro, E., Wang, D., Youn, H., & Rahwan, I. (2019). Toward

understanding the impact of artificial intelligence on labor. *Proceedings of the National*

Academy of Sciences, 116(14), 6531–6539. <https://doi.org/10.1073/pnas.1900949116>

Gawood. (2003). *Edison's Eve : a magical history of the quest for mechanical life*. Anchor

Books.

Geiselmann, R., Tsourgianni, A., Deroy, O., & Harris, L. T. (2023). Interacting with agents

without a mind: the case for artificial agents. *Current Opinion in Behavioral Sciences*, 51,

101282–101282. <https://doi.org/10.1016/j.cobeha.2023.101282>

- Giger, J.-C., Piçarra, N., Pochwatko, G., Almeida, N., Almeida, A. S., & Costa, N. (2024). Development of the Beliefs in Human Nature Uniqueness Scale and Its Associations With Perception of Social Robots. *Human Behavior and Emerging Technologies*, 2024(1), 5569587. <https://doi.org/10.1155/2024/5569587>
- Gogoll, J., & Uhl, M. (2018). Rage against the machine: Automation in the moral domain. *Journal of Behavioral and Experimental Economics*, 74, 97–103. <https://doi.org/10.1016/j.socec.2018.04.003>
- Gray, H. M., Gray, K., & Wegner, D. M. (2007). Dimensions of Mind Perception. *Science*, 315(5812), 619–619. <https://doi.org/10.1126/science.1134475>
- Gray, K., & Wegner, D. M. (2012). Feeling robots and human zombies: Mind perception and the uncanny valley. *Cognition*, 125(1), 125–130. <https://doi.org/10.1016/j.cognition.2012.06.007>
- Grossberg, S. (2020). A Path Toward Explainable AI and Autonomous Adaptive Intelligence: Deep Learning, Adaptive Resonance, and Models of Perception, Emotion, and Action. *Frontiers in Neurorobotics*, 14. <https://doi.org/10.3389/fnbot.2020.00036>
- Haslam, N., Kashima, Y., Loughnan, S., Shi, J., & Suitner, C. (2008). Subhuman, Inhuman, and Superhuman: Contrasting Humans with Nonhumans in Three Cultures. *Social Cognition*, 26(2), 248–258. <https://doi.org/10.1521/soco.2008.26.2.248>
- Hildt, E. (2019). Artificial Intelligence: Does Consciousness Matter? *Frontiers in Psychology*, 10. <https://doi.org/10.3389/fpsyg.2019.01535>
- Hooson, M. (2025, January 7). UK Artificial Intelligence (AI) Statistics And Trends In 2025. *Forbes*. <https://www.forbes.com/uk/advisor/business/software/uk-artificial-intelligence-ai-statistics/>

- Horowitz, M. C., Kahn, L., Macdonald, J., & Schneider, J. (2024). Adopting AI: How familiarity breeds both trust and contempt. *AI & SOCIETY*, 39(4), 1721–1735.
<https://doi.org/10.1007/s00146-023-01666-5>
- Huebner, B. (2010). Commonsense concepts of phenomenal consciousness: Does anyone care about functional zombies? *Phenomenology and the Cognitive Sciences*, 9(1), 133–155.
<https://doi.org/10.1007/s11097-009-9126-6>
- Hwang, A. H.-C., & Won, A. S. (2022). AI in Your Mind: Counterbalancing Perceived Agency and Experience in Human-AI Interaction. *Extended Abstracts of the 2022 CHI Conference on Human Factors in Computing Systems*, 1–10.
<https://doi.org/10.1145/3491101.3519833>
- Jacobs, O. L., Gazzaz, K., & Kingstone, A. (2022). Mind the Robot! Variation in Attributions of Mind to a Wide Set of Real and Fictional Robots. *International Journal of Social Robotics*, 14(2), 529–537. <https://doi.org/10.1007/s12369-021-00807-4>
- Joo, M. (2024). It's the AI's fault, not mine: Mind perception increases blame attribution to AI. *PLOS ONE*, 19(12), e0314559. <https://doi.org/10.1371/journal.pone.0314559>
- Kaya, F., Aydin, F., Schepman, A., Rodway, P., Yetişensoy, O., & and Demir, K.M. (2022). The Roles of Personality Traits, AI Anxiety, and Demographic Factors in Attitudes toward Artificial Intelligence. *International Journal of Human–Computer Interaction*, 40(2), 497–514. <https://doi.org/10.1080/10447318.2022.2151730>
- Kim, J. J. H., Soh, J., Kadkol, S., Solomon, I., Yeh, H., Srivatsa, A. V., Nahass, G. R., Choi, J. Y., Lee, S., Nyugen, T., & Ajilore, O. (2025). AI Anxiety: A comprehensive analysis of psychological factors and interventions. *AI and Ethics*. <https://doi.org/10.1007/s43681-025-00686-9>

- Kneer, M., & Stuart, M. T. (2021). Playing the Blame Game with Robots. *Companion of the 2021 ACM/IEEE International Conference on Human-Robot Interaction*, 407–411.
<https://doi.org/10.1145/3434074.3447202>
- Ladak, A., Loughnan, S., & Wilks, M. (2023). The Moral Psychology of Artificial Intelligence. *Current Directions in Psychological Science*, 33(1), 27–34.
<https://doi.org/10.1177/09637214231205866>
- Ladak, A., Wilks, M., Loughnan, S., & Anthis, J. R. (2025). *Robots, Chatbots, Self-Driving Cars: Perceptions of Mind and Morality Across Artificial Intelligences*.
<https://arxiv.org/pdf/2502.18683>
- Lagnado, D. A., & Channon, S. (2008). Judgments of cause and blame: The effects of intentionality and foreseeability. *Cognition*, 108(3), 754–770.
<https://doi.org/10.1016/j.cognition.2008.06.009>
- Lee, S., Lee, N., & and Sah, Y. J. (2020). Perceiving a Mind in a Chatbot: Effect of Mind Perception and Social Cues on Co-presence, Closeness, and Intention to Use. *International Journal of Human–Computer Interaction*, 36(10), 930–940.
<https://doi.org/10.1080/10447318.2019.1699748>
- Legaspi, R., Xu, W., Konishi, T., Wada, S., Kobayashi, N., Naruse, Y., & Ishikawa, Y. (2024). The sense of agency in human–AI interactions. *Knowledge-Based Systems*, 286, 111298.
<https://doi.org/10.1016/j.knosys.2023.111298>
- Liu, B. (2021). In AI We Trust? Effects of Agency Locus and Transparency on Uncertainty Reduction in Human–AI Interaction. *Journal of Computer-Mediated Communication*, 26(6), 384–402. <https://doi.org/10.1093/jcmc/zmab013>

- Liu, P., & Du, Y. (2021). Blame Attribution Asymmetry in Human–Automation Cooperation. *Risk Analysis*, 42(8), 1769–1783. <https://doi.org/10.1111/risa.13674>
- Lynch, C. R. (2021). Artificial Emotional Intelligence and the Intimate Politics of Robotic Sociality. *Space and Polity*, 25(2), 184–201. <https://doi.org/10.1080/13562576.2021.1985853>
- Madhavan, P., & Wiegmann, D. A. (2007). Similarities and differences between human–human and human–automation trust: An integrative review. *Theoretical Issues in Ergonomics Science*, 8(4), 277–301. <https://doi.org/10.1080/14639220500337708>
- Malle, B. F. (2019). How Many Dimensions of Mind Perception Really Are There? *Proceedings of the Annual Meeting of the Cognitive Science Society*, 41(0). <https://escholarship.org/uc/item/3j55p1rz>
- Malle, B. F., Guglielmo, S., & Monroe, A. E. (2014). A Theory of Blame. *Psychological Inquiry*, 25(2), 147–186. <https://doi.org/10.1080/1047840x.2014.877340>
- Mannes, A. (2020). Governance, Risk, and Artificial Intelligence. *AI Magazine*, 41(1), Article 1. <https://doi.org/10.1609/aimag.v41i1.5200>
- McMurtrie, B. L. (2023). Investigating the Dimensions of Mind Perception. *Journal of European Psychology Students*, 14(1), 54–68. <https://doi.org/10.59477/jeps.594>
- Mori, M., MacDorman, K. F., & Kageki, N. (2012). The Uncanny Valley [From the Field]. *IEEE Robotics & Automation Magazine*, 19(2), 98–100. <https://doi.org/10.1109/MRA.2012.2192811>
- Müller, B. C. N., Gao, X., Nijssen, S. R. R., & Damen, T. G. E. (2020). I, Robot: How Human Appearance and Mind Attribution Relate to the Perceived Danger of Robots.

International Journal of Social Robotics, 13(4), 691–701. <https://doi.org/10.1007/s12369-020-00663-8>

Nijssen, S. R. R., Müller, Barbara C. N., Bosse, Tibor, & Paulus, M. (2022). Can you count on a calculator? The role of agency and affect in judgments of robots as moral agents. *Human–Computer Interaction*, 38(5–6), 400–416. <https://doi.org/10.1080/07370024.2022.2080552>

Pauketat, J. V. T., & Anthis, J. R. (2022). Predicting the moral consideration of artificial intelligences. *Computers in Human Behavior*, 136, 107372. <https://doi.org/10.1016/j.chb.2022.107372>

R Core Team. (2024). *R: A Language and Environment for Statistical Computing*. R Foundation for Statistical Computing. <https://www.r-project.org/>

Rashid, A. B., & Kausik, M. A. K. (2024). AI revolutionizing industries worldwide: A comprehensive overview of its diverse applications. *Hybrid Advances*, 7, 100277. <https://doi.org/10.1016/j.hybadv.2024.100277>

Roth, E., & Davis, W. (2024, September 29). *California governor vetoes major AI safety bill*. The Verge. <https://www.theverge.com/2024/9/29/24232172/california-ai-safety-bill-1047-vetoed-gavin-newsom>

Russell, S., Norvig, P., Fabrice Popineau, Laurent Miclet, & Cadet, C. (2021). Intelligence artificielle : une approche moderne (4^e édition). *Hal.science*. <https://hal.science/hal-04245057>

Schepman, A., & Rodway, P. (2020). Initial validation of the general attitudes towards Artificial Intelligence Scale. *Computers in Human Behavior Reports*, 1. <https://doi.org/10.1016/j.chbr.2020.100014>

- Schoenherr, J. R. & Thomson, R. (2024). When AI Fails, Who Do We Blame? Attributing Responsibility in Human-AI Interactions. *IEEE Transactions on Technology and Society*, 1–1. <https://doi.org/10.1109/tts.2024.3370095>
- Shabbir, J., & Anwer, T. (2018). Artificial Intelligence and its Role in Near Future. *ArXiv (Cornell University)*, 14(8). <https://doi.org/10.48550/arxiv.1804.01396>
- Shank, D. B., & DeSanti, A. (2018). Attributions of morality and mind to artificial intelligence after real-world moral violations. *Computers in Human Behavior*, 86, 401–411. <https://doi.org/10.1016/j.chb.2018.05.014>
- Shank, D. B., North, M., Arnold, C., & Gamez, P. (2021). Can mind perception explain virtuous character judgments of artificial intelligence? *Technology, Mind, and Behavior*, 2(2). <https://doi.org/10.1037/tmb0000047>
- Srinivasan, R., & Sarial-Abi, G. (2021). When Algorithms Fail: Consumers' Responses to Brand Harm Crises Caused by Algorithm Errors. *Journal of Marketing*, 85(5), 74–91. <https://doi.org/10.1177/0022242921997082>
- Stafford, R. Q., MacDonald, B. A., Jayawardena, C., Wegner, D. M., & Broadbent, E. (2013). Does the Robot Have a Mind? Mind Perception and Attitudes Towards Robots Predict Use of an Eldercare Robot. *International Journal of Social Robotics*, 6(1), 17–32. <https://doi.org/10.1007/s12369-013-0186-y>
- Statement on AI Risk | CAIS*. (n.d.). [www.safe.ai](https://www.safe.ai/work/statement-on-ai-risk). <https://www.safe.ai/work/statement-on-ai-risk>
- Sullivan, Y., de Bourmont, M., & Dunaway, M. (2022). Appraisals of harms and injustice trigger an eerie feeling that decreases trust in artificial intelligence systems. *Annals of Operations Research*, 308(1), 525–548. <https://doi.org/10.1007/s10479-020-03702-9>

- Tanibe, T., Hashimoto, T., & Karasawa, K. (2017). We perceive a mind in a robot when we help it. *PLOS ONE*, 12(7), e0180952. <https://doi.org/10.1371/journal.pone.0180952>
- Troshani, I., Rao Hill, S., Sherman, C., & Arthur, D. (2020). Do We Trust in AI? Role of Anthropomorphism and Intelligence. *Journal of Computer Information Systems*, 61(5), 1–11. <https://doi.org/10.1080/08874417.2020.1788473>
- van der Woerdt, S., & Haselager, P. (2019). When robots appear to have a mind: The human perception of machine agency and responsibility. *New Ideas in Psychology*, 54, 93–100. <https://doi.org/10.1016/j.newideapsych.2017.11.001>
- Vanneste, B. S., & Puranam, P. (2024). Artificial Intelligence, Trust, and Perceptions of Agency. *Academy of Management Review*. <https://doi.org/10.5465/amr.2022.0041>
- Vincent, L., & Gannon, J. (2008). *Differentiating Competence, Capability and Capacity*, from <https://innovationsthatwork.com/images/pdf/June08newsltr.pdf>
- Vuong, Q., La, V., Nguyen, M., Nguyen Quang-Loc, Jin, R., La, M.-K., & Le, T.-T. (2023). *AI in pain: How knowledge influences people's perceptions of AI's capability of feeling distressed*. <https://doi.org/10.31219/osf.io/ypgxf>
- Wang, X., & Krumhuber, E. G. (2018). Mind Perception of Robots Varies With Their Economic Versus Social Function. *Frontiers in Psychology*, 9. <https://doi.org/10.3389/fpsyg.2018.01230>
- Waytz, A., Cacioppo, J., & Epley, N. (2010). Who Sees Human? The Stability and Importance of Individual Differences in Anthropomorphism. *Perspectives on Psychological Science*, 5(3), 219–232. <https://doi.org/10.1177/1745691610369336>

- Waytz, A., Heafner, J., & Epley, N. (2014). The mind in the machine: Anthropomorphism increases trust in an autonomous vehicle. *Journal of Experimental Social Psychology*, 52, 113–117. <https://doi.org/10.1016/j.jesp.2014.01.005>
- Weisman, K., Dweck, C. S., & Markman, E. M. (2017). Rethinking people’s conceptions of mental life. *PNAS Proceedings of the National Academy of Sciences of the United States of America*, 114(43), 11374–11379. <https://doi.org/10.1073/pnas.1704347114>
- Weisman, W. D., & Peña, J. F. (2021). Face the Uncanny: The Effects of Doppelganger Talking Head Avatars on Affect-Based Trust Toward Artificial Intelligence Technology are Mediated by Uncanny Valley Perceptions. *Cyberpsychology, Behavior, and Social Networking*, 24(3), 182–187. <https://doi.org/10.1089/cyber.2020.0175>
- Wilks, M., Basarkod, G., & Gray, K. (In prep). Attitudes towards artificial intelligence are politically neutral.
- Xu, X., & Sar, S. (2018). *Do We See Machines The Same Way As We See Humans? A Survey On Mind Perception Of Machines And Human Beings*. 472–475. <https://doi.org/10.1109/roman.2018.8525586>
- Xu, Y., Wang, Q., An, Z., Wang, F., Zhang, L., Wu, Y., Dong, F., Qiu, C.-W., Liu, X., Qiu, J., Hua, K., Su, W., Xu, H., Han, Y., Cao, X., Liu, E., Fu, C., Yin, Z., Liu, M., & Roepman, R. (2021). Artificial Intelligence: a Powerful Paradigm for Scientific Research. *The Innovation*, 2(4). Sciencedirect. <https://doi.org/10.1016/j.xinn.2021.100179>
- Yampolskiy, R. V. (2018). Predicting future AI failures from historic examples. *Foresight*, 21(1). <https://doi.org/10.1108/fs-04-2018-0034>

- Young, A. D., & Monroe, A. E. (2019). Autonomous morals: Inferences of mind predict acceptance of AI behavior in sacrificial moral dilemmas. *Journal of Experimental Social Psychology*, 85, 103870. <https://doi.org/10.1016/j.jesp.2019.103870>
- Yudkowsky, E. (2008). Artificial Intelligence as a positive and negative factor in global risk. In E. Yudkowsky, *Global Catastrophic Risks*. Oxford University Press.
<https://doi.org/10.1093/oso/9780198570509.003.0021>
- Złotowski, J., Yogeeswaran, K., & Bartneck, C. (2017). Can we control it? Autonomous robots threaten human identity, uniqueness, safety, and resources. *International Journal of Human-Computer Studies*, 100, 48–54. <https://doi.org/10.1016/j.ijhcs.2016.12.008>

Appendices

Appendix A

Table A1

Sample composition by gender, education, and ethnicity

Variable	Value	Count	Percentage
Gender			
	Male	46	21.4
	Female	158	73.5
	Other	11	5.1
Education			
	High School	121	56.3
	Some College	44	20.5
	Associate Degree	2	0.9
	Bachelor's Degree	31	14.4
	Master's Degree	14	6.5
	Professional Degree or Doctorate Degree	3	1.4
Ethnicity			
	African	1	0.5
	Arab	2	0.9
	Asian	83	38.6
	Mixed or Multiple Ethnic Groups	19	8.8
	Other	5	2.3
	White	105	48.8

Table A2

Sample composition by age and political orientation

Variable	Mean	SD	Range
Age	22.29	8.62	18 - 56
Political Orientation^a	3.38	1.59	1 - 9

Note. ^a Measured on a scale of 1 (Liberal) to 9 (Conservative).

AI Safety Concern Measures

“In this survey we use the term **Artificial Intelligence (AI)** to refer to intelligent entities built by humans, such as robots, virtual copies of human brains, or computer programs that solve problems, with or without a physical body that may exist now or in the future.”

“Please rate how much you agree with the following statements on a scale of 1 (strongly disagree) to 7 (strongly agree)”

[illegible]

Appendix C

Mind Perception Scale from Gray et al. (2007)

Table B1

Factor loadings of mental capacities

Factor		
Mental Capacity	Experience	Agency
Hunger	.97	.15
Fear	.93	.31
Pain	.89	.41
Pleasure	.85	.51
Rage	.78	.58
Desire	.76	.64
Personality	.72	.68
Consciousness	.71	.69
Pride	.71	.69
Embarrassment	.70	.65
Joy	.67	.60
Self-control	.18	.97
Morality	.35	.93
Memory	.33	.91
Emotion recognition	.54	.83
Planning	.55	.82
Communication	.66	.74
Thought	.68	.73

Mind Perception Measures

“As a reminder, **Artificial Intelligence (AI)** refers to intelligent entities built by humans, such as robots, virtual copies of human brains, or computer programs that solve problems, with or without a physical body that may exist now or in the future.”

“How much do you agree or disagree that **current** AIs are capable of the following capacities on a scale of Strongly disagree (1) to Strongly agree (7)?”

[illegible]

Scale prompt:

“How much do you agree or disagree that **future** AIs are capable of the following capacities on a scale of Strongly disagree (1) to Strongly agree (7)?”

[illegible]

Appendix E

Attention Checks

First attention check

Scale prompt:

“Please rate how much you agree with the following statements on a scale of 1 (strongly disagree) to 7 (strongly agree). You can work quickly; your first feeling is generally best.”

	Strongly disagree (1)	Disagree (2)	Somewhat disagree (3)	Neither agree nor disagree (4)	Somewhat agree (5)	Agree (6)	Strongly agree (7)
Please select somewhat agree (5) to show that you are paying attention	☐	☐	☐	☐	☐	☐	☐

Second attention check

What did we ask you about today?

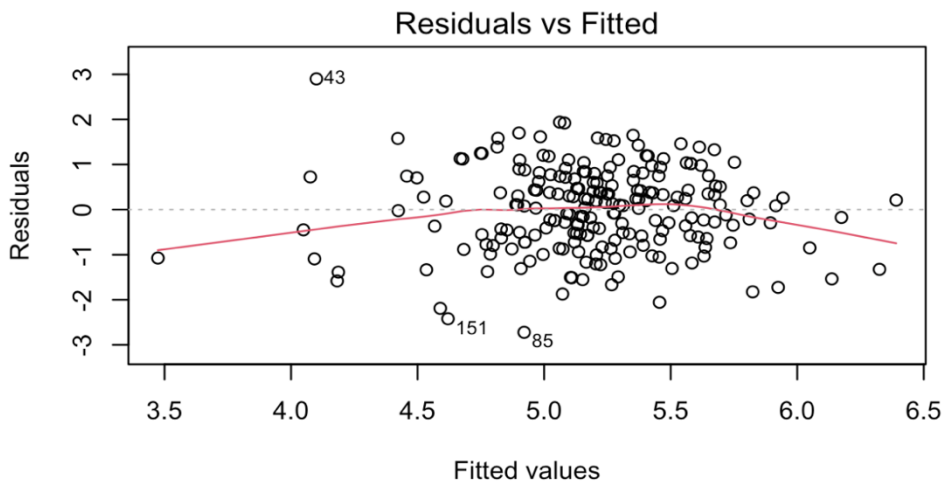
- ☐ Beliefs about artificial intelligence
- ☐ Beliefs about climate change
- ☐ Beliefs about space travel

Appendix F

Assumption Checks

Figure F1

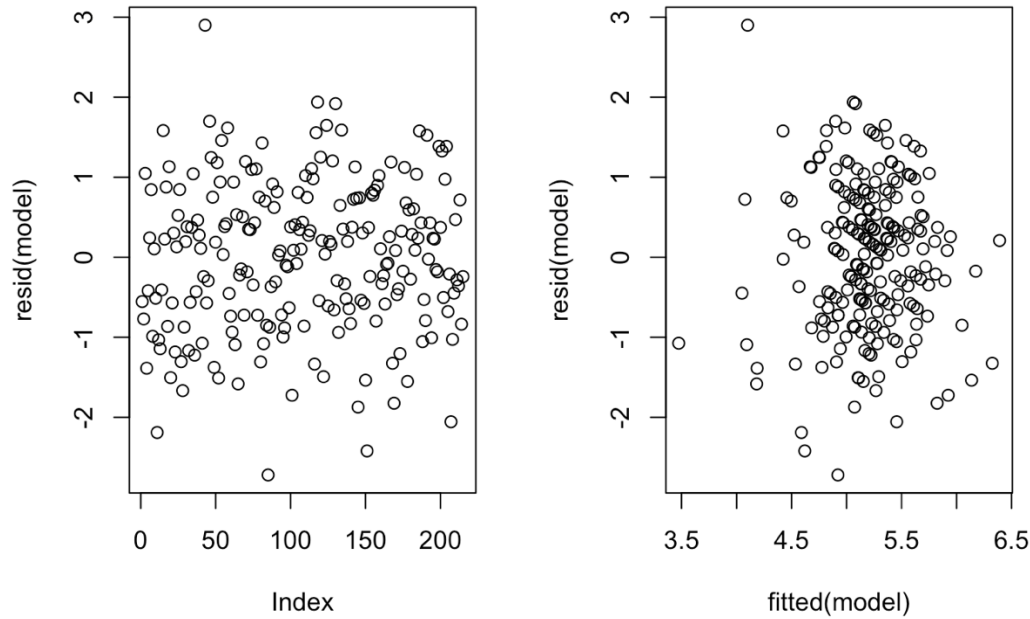
Linearity: Residuals vs. Fitted plot



The Residuals vs. Fitted plot displayed that residuals were randomly scattered with a mean of zero, indicating that the assumption of linearity was met.

Figure F2

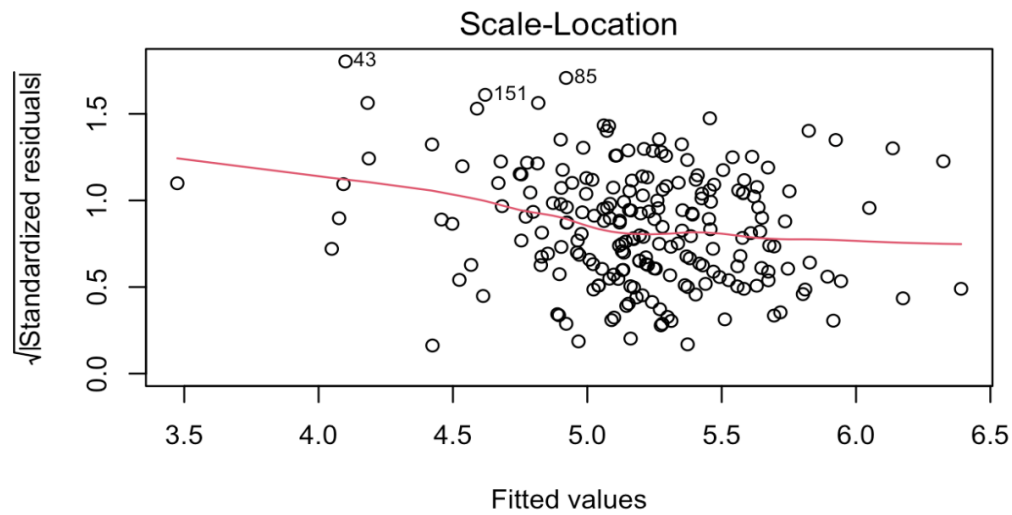
Independence of errors: Residuals vs. Index plot



The independence of errors assumption was met, as the data were collected via an online survey where each participant responded individually. Additionally, a Residuals vs. Index plot showed no discernible patterns, further supporting this assumption.

Figure F3

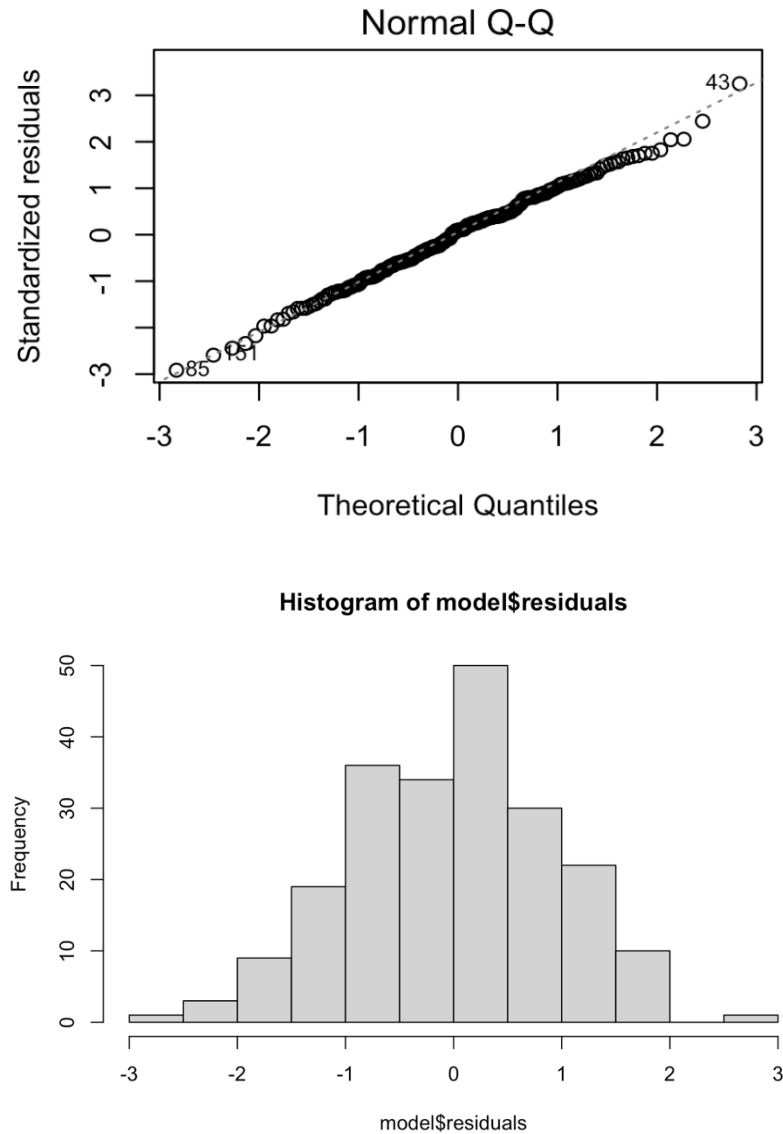
Homoscedasticity: Scale-Location plot



The Scale-Location plot displayed a relatively flat trend line and an even spread of residuals across the range of fitted values, indicating that the assumption of homoscedasticity was met.

Figure F4

Normality of residuals: Q-Q plot and histogram of residuals



The Q-Q plot displayed that residuals lined well on the dashed diagonal line, ensuring normality of the model. This is further supported by the histogram of residuals.

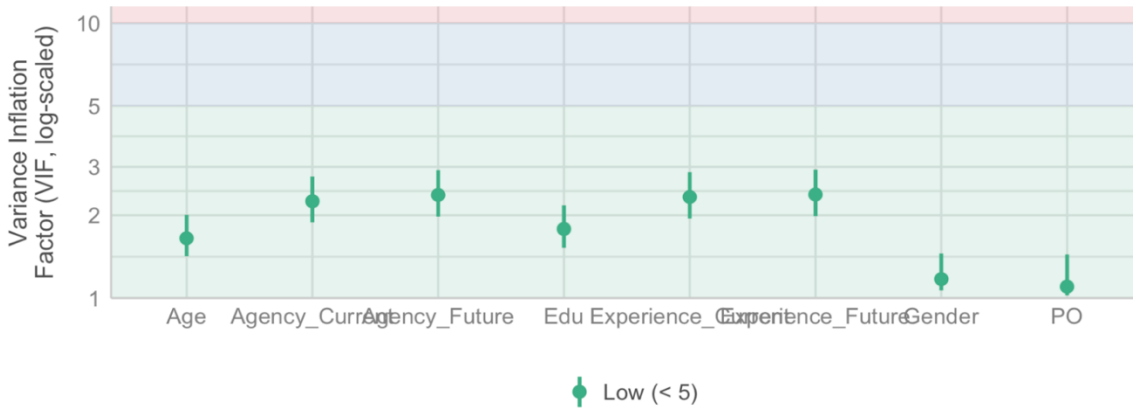
Figure F5

Multicollinearity

	GVIF	Df	GVIF^{1/(2*Df)}	Interacts With	Other Predictors
Agency_Current	2.25	1	1.50	--	Agency_Future, Experience_Current, Experience_Future, Age, Gender, PO, Edu
Agency_Future	2.37	1	1.54	--	Agency_Current, Experience_Current, Experience_Future, Age, Gender, PO, Edu
Experience_Current	2.33	1	1.53	--	Agency_Current, Agency_Future, Experience_Future, Age, Gender, PO, Edu
Experience_Future	2.38	1	1.54	--	Agency_Current, Agency_Future, Experience_Current, Age, Gender, PO, Edu
Age	1.65	1	1.28	--	Agency_Current, Agency_Future, Experience_Current, Experience_Future, Gender, PO, Edu
Gender	1.17	2	1.04	--	Agency_Current, Agency_Future, Experience_Current, Experience_Future, Age, PO, Edu
PO	1.10	1	1.05	--	Agency_Current, Agency_Future, Experience_Current, Experience_Future, Age, Gender, Edu
Edu	1.78	5	1.06	--	Agency_Current, Agency_Future, Experience_Current, Experience_Future, Age, Gender, PO

Collinearity

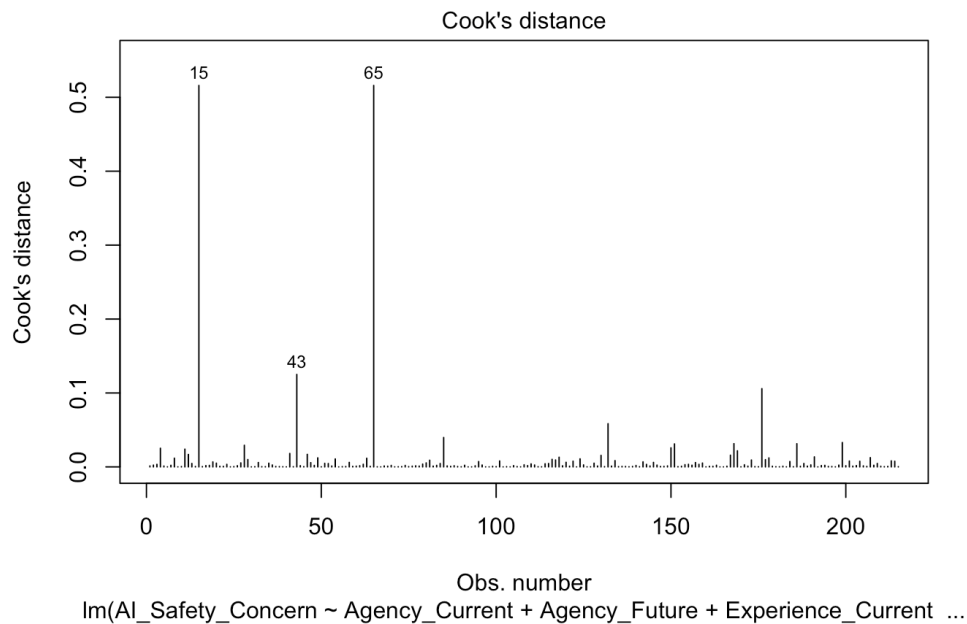
High collinearity (VIF) may inflate parameter uncertainty



VIF values for all predictors were less than 5, indicating that multicollinearity was not adversely affecting the model estimates.

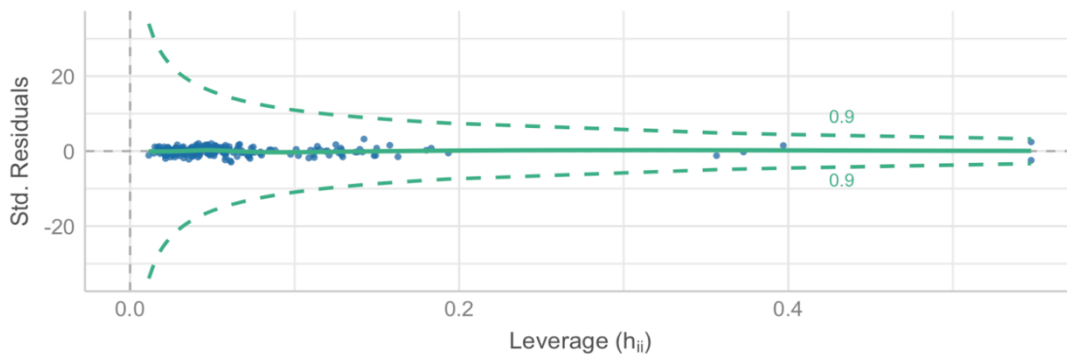
Figure F6

Cases of high influence: Cook's Distance



Influential Observations

Points should be inside the contour lines



Observations were assessed using Cook's Distance, where a value >1 would be considered to have a high influence. The plot indicated that no observations have a high influence on the model estimates. This was further supported by the Residuals vs. Leverage plot.